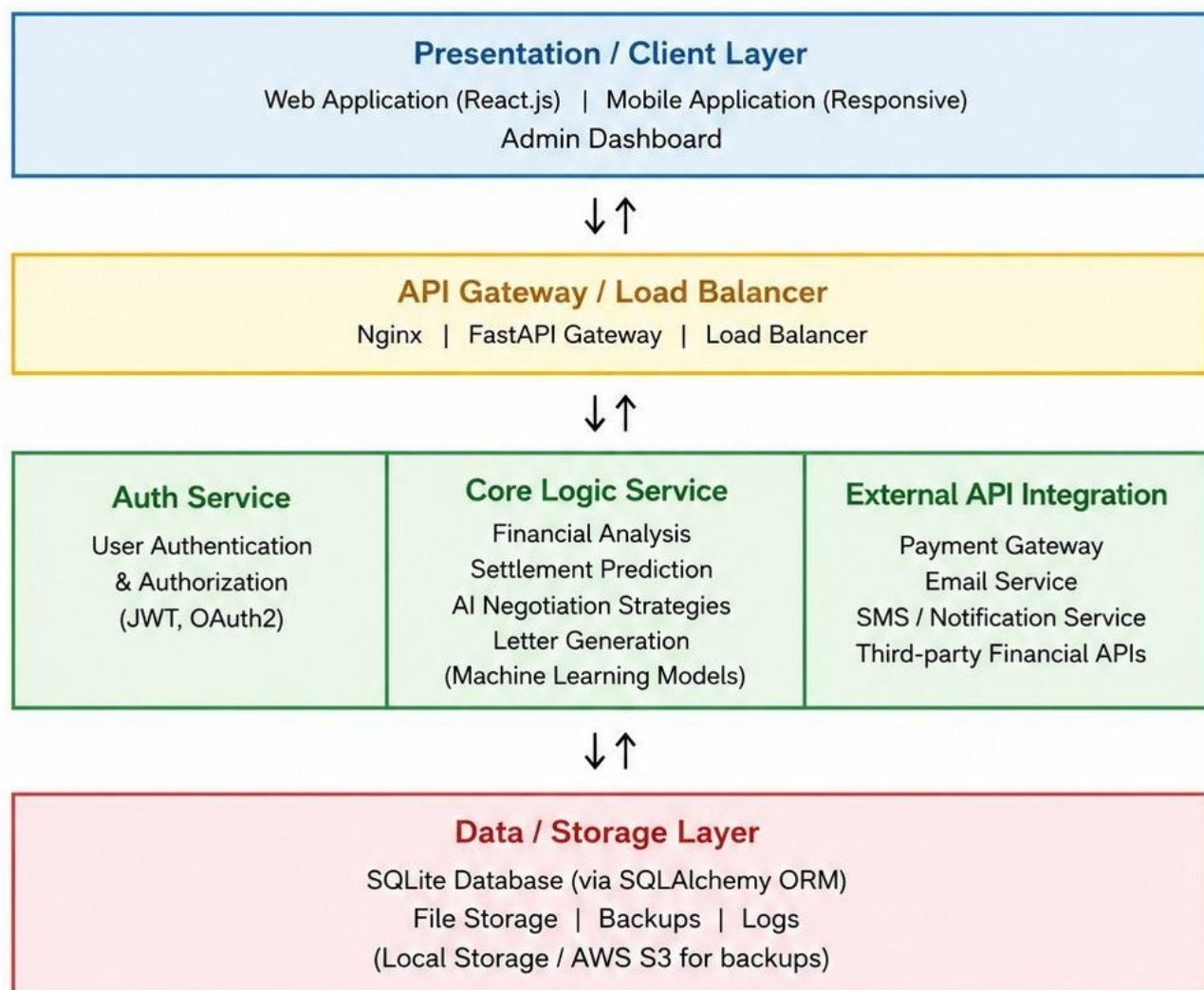


Solution Architecture

Date	5 July 2026
Team ID	SWTID-2026-5455
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below illustrates the high-level architecture of the AI Powered Debt Relief & Financial Recovery Platform. It shows the flow and interaction between the Presentation Layer, Application Layer, and Data Layer along with key services and technologies used.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for borrowers to interact with the platform. Includes web and mobile applications and admin dashboard.	React.js, HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Acts as a single entry point for all client requests, handles routing, load balancing, and request validation.	Nginx, FastAPI Gateway, Uvicorn
Microservices	Contains core business services such as authentication, financial analysis, settlement prediction, letter generation, and notifications.	Python (FastAPI), SQLAlchemy, Pydantic, Celery
Database	Stores all application data including users, loans, settlements, AI analysis results, and system logs.	SQLite, SQLAlchemy ORM